Nowcasting

Workbook

Workshop 4 – Nowcasting using Standard Nowcasting Approach in EViews: India Case Study



IMF Institute for Capacity Development

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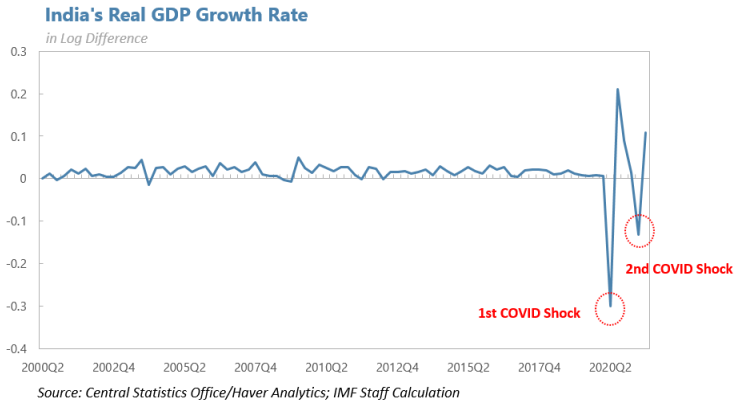
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Notes on India's Economy

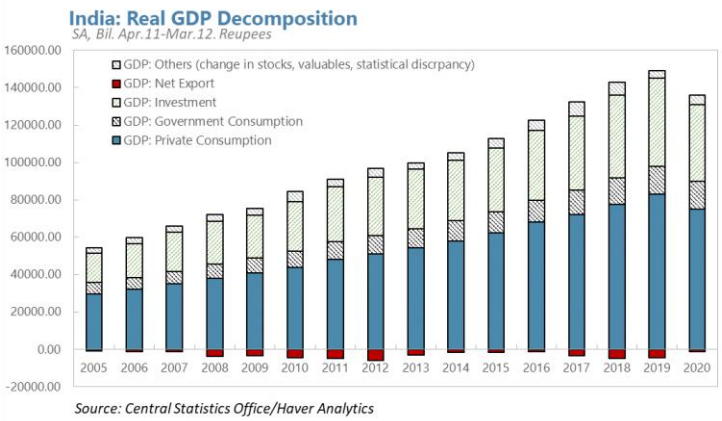
India is bounded by the Indian Ocean on the south and the Arabian Sea on the southwest. In addition, it shares land borders with Pakistan, China, Nepal, Bhutan, Bangladesh, and Myanmar. In 2020, India was ranked as the 6th largest economy by GDP (in current US$). Before COVID-19, India was among the fastest-growing economies in the world.



Source: Central Statistics Office/Haver Analytics; IMF Staff Calculation

According to the IMF Executive Board Article IV consultation with India in October 2021[1], India experienced two COVID-19 waves. The first caused an unprecedented contraction in real GDP of 7.3 percent in the fiscal year 2021. The second resulted in a smaller and shorter fall in activity. The combined impact of pandemic-related support and the contraction in economic activity led to a widening of the fiscal deficit to 12.8 percent of GDP.

The economic outlook for India remains unclear given pandemic-related uncertainties. Medium-term growth will be likely be impacted by the adverse shocks in investment, human capital, and other growth factors. However, more efficient vaccination and better therapeutics should reduce the negative long-term impact of the pandemic.

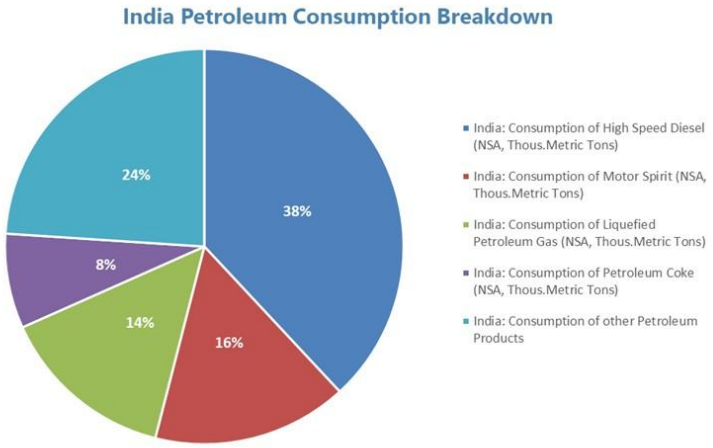


Source: Central Statistics Office/Haver Analytics

India's Gross Domestic Product is highly dependent on private final consumption expenditures, accounting for more than 50 percent of the real GDP in 2020. Food and nonalcoholic beverages have the

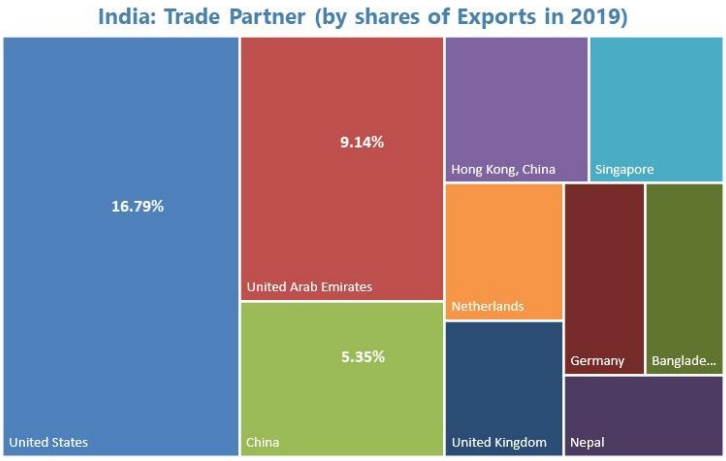
[1] Source: IMF, IMF Executive Board Concludes 2021 Article IV Consultation with India

most weight in private consumption, accounting for 27 percent of the real private consumption in 2019.[2] Transport has the second highest weight in private consumption, accounting for 18 percent of the real private consumption in 2020.



Source: Ministry of Petroleum & Natural Gas/Haver Analytics

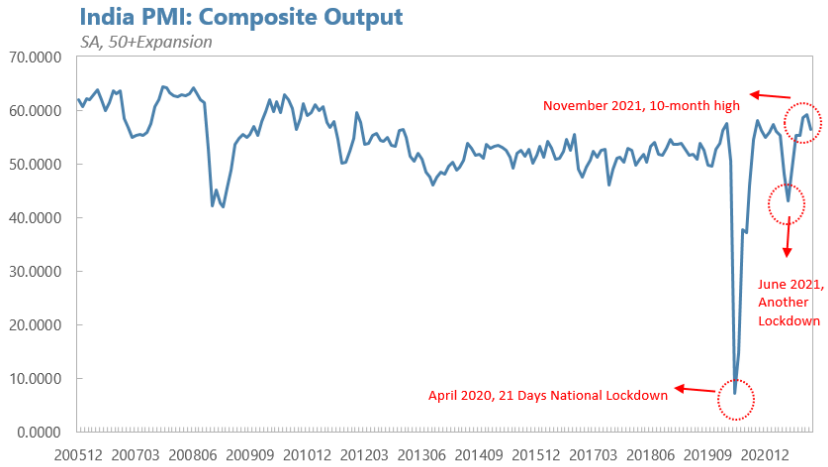
India's three largest trade partners (ranked from largest to smallest) are the United States, United Arab Emirates, and China. Consumer goods take the most weight among the export products, accounting for 45 percent of exports. Among the import products, raw materials are at the top, accounting for 33 percent.[3]



Source: WBG WITS Data

Since the pandemic, India has experienced significant drops in the Purchasing Manager Index (PMI). The PMI dropped to a historical low of 7.1 in April 2020, reflecting impact of 21 days of national lockdown that began March 24th. However, the PMI recovered gradually after the easing of the lockdown, exceeding the neutral threshold (i.e., 50.0) in September. In November 2021, the PMI reached a ten-month high of 59.2, reflecting a solid expansion in the manufacturing sector.

[2] Based on data provided by the Central Statistics Office.
[3] According to World Bank Group's WITS (India Trade Summary 2019 | WITS Data (worldbank.org))



Source: Caixin/IHS Markit

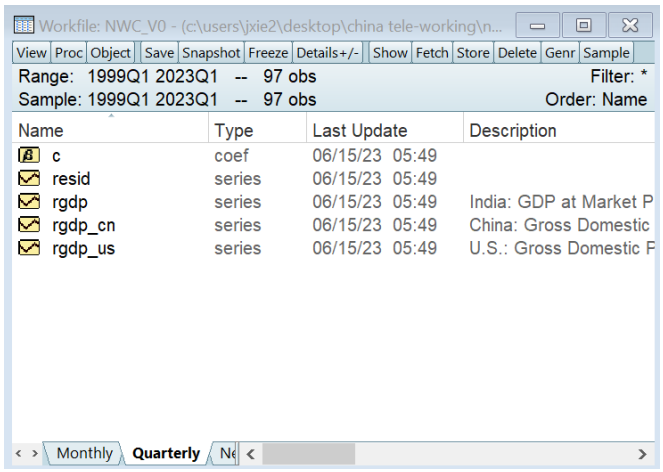
Workshop Objective

The aim of this case study is to extend India's quarterly real GDP data, which at the time of writing was available till the first quarter of 2023, to the second quarter of **2023**. We shall use higher frequency data (in this case, monthly) in various ways to do so.

Dataset

The selected data relevant to India's macroeconomy real growth can be found in the EViews workfile **nwc.wf1**. The workfile has two "pagefiles" initially – one for each frequency: **Quarterly** and **Monthly**. Additional pagefiles will be added to the workfile as we generate nowcasts using different estimators (e.g., Bridge, MIDAS, Machine Learning methods).

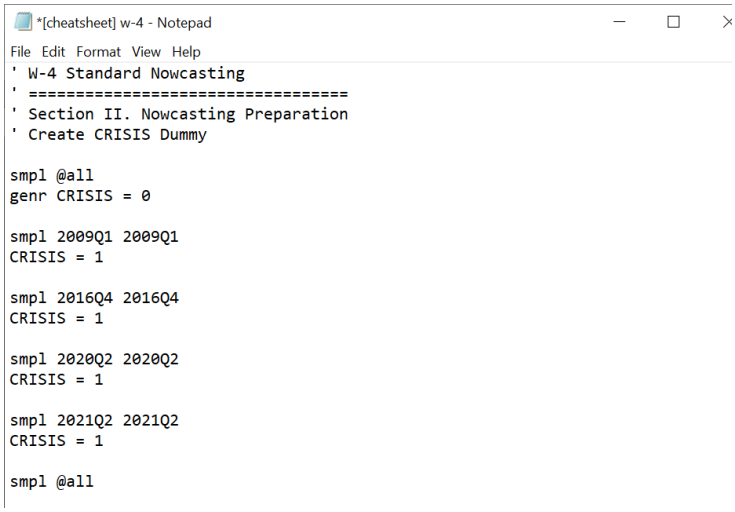
Figure 1: EViews Workfile for India ("nwc.wf1")



Section 0: EViews Command Cheat Sheet

You may refer to the text file ***[cheatsheet] w-4.rtf*** to complete this workshop. It contains most of the commands used in the workshop. You can avoid syntax errors by copying and pasting the specific command lines rather than typing them.

Open the text file in EViews by clicking "File => Open => Text File". Alternatively, you could use Windows Notepad (as below) to access the file outside of EViews.



Section I: Baseline Model

Figure 2 shows annual real GDP growth for India for 1999Q1-2023Q1 together with an estimate of India's long-run real GDP potential growth (orange line), which was obtained by filtering (or smoothing) the original series. Figure 3 shows the corresponding levels of real GDP (blue line) and potential output (orange line).

The volatility of annual real GDP growth in India is quite apparent, and there appears to be a slight secular decline in long-term real GDP growth.

The two charts are available in "**nwc.wf1**" as graph objects (see **graph01_rgdp_g** and **graph02_rgdp**) in the Quarterly pagefile. You may generate these two charts following these steps:

1. Activate the "Quarterly" pagefile, and then select the series containing the real GDP data (**rgdp)**.
2. Generate the trend of real GDP by clicking "Proc" => "Hodrick-Prescott Filter" => "OK". The HP filtered series will be named "**hptrend01**"
3. Select **rgdp** and **hptrend01** and then generate a graph.
4. Apply the same process to the real GDP growth rate (you may generate **rgdp_growth** using the command: **series rgdp_growth = log(rgdp)-log(rgdp(-4))**.

Figure 2: Real GDP growth and Potential Growth, India

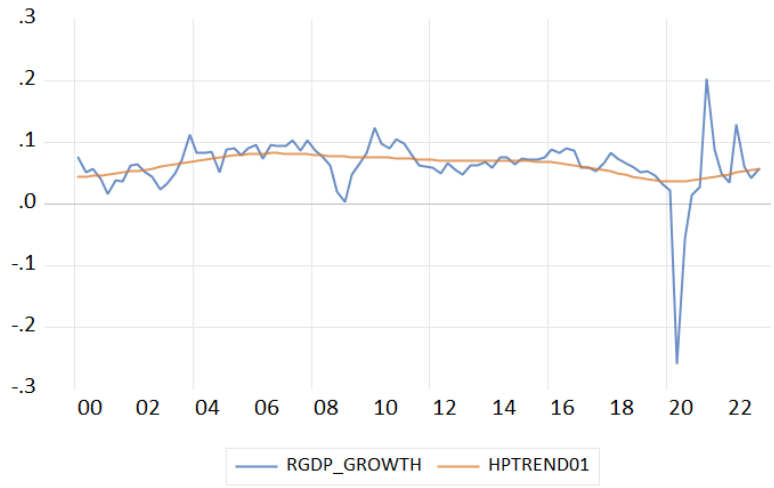
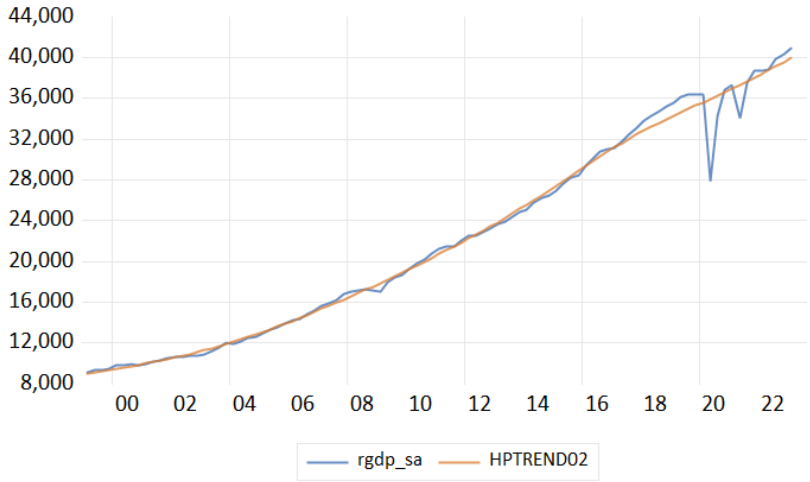


Figure 3: Real GDP and Potential Output, India



Our objective is to extend India's real GDP data to **2023Q2** using higher frequency data (i.e., monthly) that is published or been extended (via some forecasting procedure) to at least **2023Q2**.

We will use 2 monthly series and 2 quarterly series[4] to do so, all of which have been updated after the latest real GDP was released. All these series are likely to be leading indicators of India's real GDP growth:

- **Consumption of Motor Spirit (Thousands. Metric Tons)**, available until 2023M05. It is "**consu_motor**" in the "Monthly" pagefile. Motor spirit is a generic name for automotive fuels

[4] **consu_motor** (monthly) series was collected from India's Ministry of Petroleum & Natural Gas. **pmi_new** (monthly) series was collected from IHS Markit. **rgdp_cn** (quarterly) was collected from China's National Bureau of Statistics. **rgdp_us** (quarterly) was collected from US Bureau of Economic Analysis, and **rgdp** (quarterly) was collected from India's Central Statistics Office.

and is often known as "Gasoline" in North America. According to India's Ministry of Petroleum and Natural Gas, motor spirit ranked as the second-highest consumption of petroleum products, accounting for 15% of India's total Petroleum Products in October 2021. Several papers have confirmed the key relationship between petroleum consumption and economic growth.[5] Theoretically, an increase in gasoline consumption implies growth in the transportation, industrial, and household sectors.

- **Purchasing Manager Index (PMI, SA, 50+ = Expansion)**, available until 2023M05. It is "**pmi**" in the "Monthly" pagefile. PMI is an important leading indicator measuring economic trends in the manufacturing and service sectors. The composite PMI ranges from 0 to 100. Above 50 indicates an expansion compared to the previous period. A PMI increase should be positively correlated with a growth in the real GDP.

- **United States' Real Gross Domestic Product (SA, in Billions, in 2012 dollars),** available until 2023Q1. It is "**rgdp_us**" in the "Quarterly" pagefile. The United States and India have maintained a strong political and economic relationship. The United States is the second-largest investor for India's Foreign Direct Investment, amounting to 102.45 billion Indian rupees in 2021.[6] Meanwhile, the United States is India's largest trade partner, accounting for 16.79 percent of exports in 2019, according to World Bank's World Integrated Trade Solution (WITS). Thus, there is likely to be a positive relationship between these two countries' economic growth.

- **China's Real Gross Domestic Product (SA, in Billion, 2020 Yuan)**, available until 2023Q1. It is "**rgdp_cn**" in the "Quarterly" pagefile. Since the beginning of this century, the bilateral trade between India and China has increased significantly. The total trade between the two countries in 2021 reached at USD 125.66 billion, up 43.3 percent from 2020.[7] According to the WITS, China was placed as India's third largest trade partner, right after the United States and the United Arab Emirates. The two countries' economic growth is likely to be positively related.

The baseline forecasting model will be an auto-regressive distributed lag model for quarterly GDP growth estimated in log differences. It includes 3 lags of the first difference of log real GDP, together with quarterly versions of **consu_motor** and **pmi_new**. Also, one lag was used to allow for the delayed impact of **rgdp_us** and **rgdp_cn** on **rgdp**. The base model also has a dummy variable to capture (proxy for) the negative effects of financial crisis, 2016 demonetization, and 2 COVID-19 waves (i.e., **crisis**, which is in the quarterly pagefile).[8] The differencing order of the variables was chosen to ensure that the variables in the regression are stationary. Nonstationary series were differenced to make them stationary.

[5] Habib MU. An Econometric Analysis of Relationship between Economic Growth and Petroleum Consumption in India. Jindal Journal of Business Research. 2019;8(2):118-127. doi:10.1177/2278682119833195

[6] Source: https://www.statista.com/statistics/1020989/india-fdi-equity-inflows-investing-countries/

[7] Source: China's General Administration of Customs (GAC); https://economictimes.indiatimes.com/news/economy/foreign-trade/india-china-trade-grows-to-record-125-billion-in-2021-despite-tensions-in-eastern-ladakh/articleshow/88900383.cms?from=mdr

[8] The base model should be seen as an illustrative statement about the determinants of real GDP growth in India and is subject to change (e.g., one may prefer to use lagged versions of the indicators given that they are both

In terms of the EViews command language, the base regression to be run using quarterly data is:

dlog(rgdp) dlog(rgdp(-1)) dlog(rgdp(-2)) dlog(rgdp(-3)) dlog(consu_motor) d(pmi_new) dlog(rgdp_us(-1)) dlog(rgdp_cn(-1)) crisis c

We may determine the feasible sample range for the regression by checking the start and end date of all the series available.

***Note**: we selected the baseline model based on three criteria:*

1. **Akaike Info Criterion (AIC)**: used for model selection; models with smaller AIC values are preferred. AIC is the preferred statistic for finding the best forecasting model.

2. **Adjusted R-Squared**: Measures the success of the regression in predicting the values of dependent variable (percent variation in the dependent variable explained, adjusting for degrees of freedom). Adjusts for the number of independent regressors by penalizing R-squared for additional regressors. Larger values are preferred.

3. **Root Mean Squared Error (RMSE)**: RMSE is the square root of the residual sum of squares. Smaller values are preferred. It can be applied within sample by running the regression on a subset of the available data and using it to forecast the non-included observations.

leading indicators of GDP growth). Moreover, the base model is not a structural model. Feel free to experiment with alternative specifications to improve the accuracy of the nowcast.

Baseline Option 1

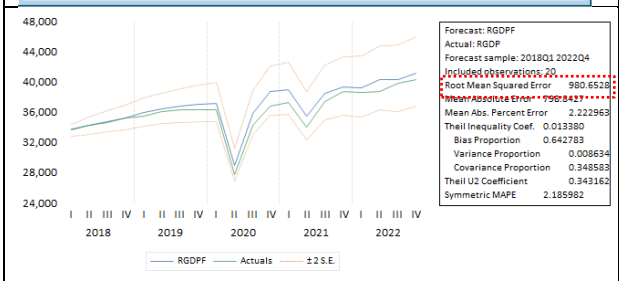
Equation: UNTITLED Workfile: NWC_V2_BRIDGE::Quarterly\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: DLOG(RGDP)
Method: Least Squares
Date: 10/10/23 Time: 03:24
Sample (adjusted): 2000Q2 2022Q4
Included observations: 91 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DLOG(RGDP(-3))	-0.017218	0.038781	-0.443990	0.6582
DLOG(RGDP(-2))	-0.041545	0.040685	-1.021145	0.3102
DLOG(RGDP(-1))	-0.032739	0.065945	-0.496464	0.6209
DLOG(CONSU_MOTOR)	0.312479	0.036245	8.621324	0.0000
D(PMI_NEW)	0.001660	0.000405	4.101695	0.0001
DLOG(RGDP_US(-1))	0.091137	0.167486	0.544148	0.5878
DLOG(RGDP_CN(-1))	0.381022	0.123913	3.074913	0.0029
CRISIS	-0.022380	0.007359	-3.041240	0.0032
C	0.003584	0.002842	1.261179	0.2108

R-squared	0.926241	Mean dependent var	0.015606
Adjusted R-squared	0.919045	S.D. dependent var	0.040663
S.E. of regression	0.011570	Akaike info criterion	-5.987209
Sum squared resid	0.010976	Schwarz criterion	-5.738882
Log likelihood	281.4180	Hannan-Quinn criter.	-5.887025
F-statistic	128.7158	Durbin-Watson stat	2.214563
Prob(F-statistic)	0.000000		



Baseline Option 2

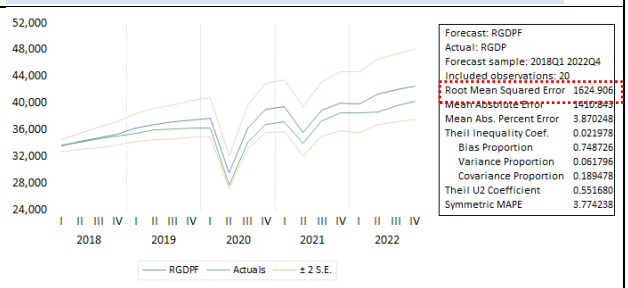
Equation: EQ_F_BASE01 Workfile: NWC_3_MIDAS::Midas\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: DLOG(RGDP)
Method: Least Squares
Date: 08/27/26 Time: 17:48
Sample (adjusted): 2000Q2 2022Q4
Included observations: 91 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DLOG(RGDP(-3))	0.008721	0.037354	0.233473	0.8160
DLOG(RGDP(-2))	0.003718	0.039528	0.094071	0.9253
DLOG(RGDP(-1))	0.007540	0.042202	0.178660	0.8586
DLOG(CONSU_MOTOR)	0.362878	0.033633	10.78929	0.0000
D(PMI_NEW)	0.002200	0.000376	5.844563	0.0000
CRISIS	-0.025008	0.007483	-3.341784	0.0012
C	0.008969	0.002296	3.906463	0.0002

R-squared	0.917658	Mean dependent var	0.015606
Adjusted R-squared	0.911776	S.D. dependent var	0.040663
S.E. of regression	0.012078	Akaike info criterion	-5.921086
Sum squared resid	0.012253	Schwarz criterion	-5.727943
Log likelihood	276.4094	Hannan-Quinn criter.	-5.843165
F-statistic	156.0221	Durbin-Watson stat	2.268975
Prob(F-statistic)	0.000000		



We will use Baseline 1 to generate our nowcasts based on lower RMSE, lower AIC value, and higher adjusted R-squared.

Section II: Data Preparation

Because the monthly versions of all the high-frequency indicators end in 2023M05, one can derive a quarterly series of these indicators based on actual data that ends in 2023Q1. This will allow a within-sample test of the baseline model to forecast real GDP by comparing it with the actual data for 2023Q1.

The first step in the data preparation phase is to remove within-sample missing values from the high-frequency data using an appropriate interpolation method. EViews supports several interpolation methods, including cubic-spline and the Chow-Lin procedures. It is important to interpolate the high-frequency data before it is seasonally adjusted or converted to the target frequency to avoid the propagation of missing values at the target frequency.

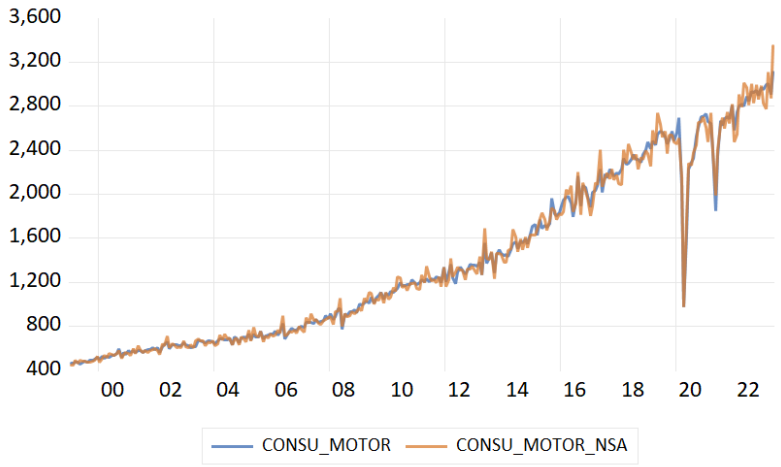
The second step is to seasonally adjust the high-frequency data if required. Note that **PMI_NEW** has already been seasonally adjusted by the data provider, but **CONSU_MOTOR** requires seasonal adjustment using the X-13 procedure. The following steps are required to do so:

Step 0: Initiate the seasonal adjustment procedure for **CONSU_MOTOR** in monthly pagefile, using the following command:

```
consu_motor.x13(save="d11", tf=auto, outtype="tc", outspan="2020.1 , 2021.4")
@x11arima(amdl=b) @reg(regs="const") @x11()
```

The seasonally adjusted data will be saved in a series called **consu_motor_d11** in the monthly page file.

Figure 4: Seasonal Adjustment of Consu_Motor (consu_motor_d11)



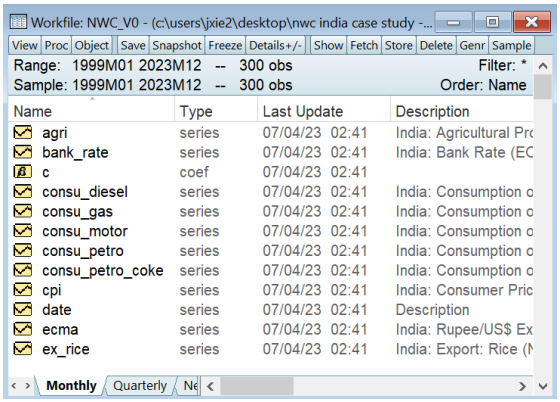
Firstly, we need to convert the monthly indicators to the target frequency (quarterly). Conversion proceeds by copying the monthly series to the **quarterly** pagefile, and then invoking "**Paste Special"** to specify the conversion method to be used to move to the quarterly frequency.

Before we can do so, however, we need to decide on a conversion method for each series to move from a higher to a lower frequency. Consider how this works for **pmi_new**. We will use "average" for **pmi_new** because it is "stock" indicator. We will use "sum" for **consu_motor** because it is "flow" indicator.[9]

Step 1: Make "Monthly" the active page file (Figure 5).

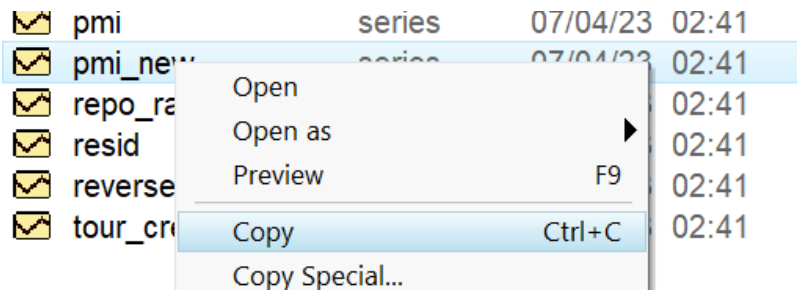
[9] The "Average" setting will compute the average of the high frequency observations for a given low frequency observations. The "Sum" setting will compute the sum of the high frequency observations for a given low frequency observations.

Figure 5: EViews Workfile with "Monthly" the active pagefile.



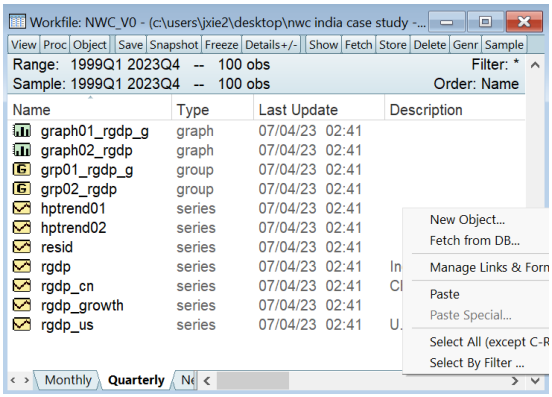
Step 2: Select **pmi_new**, right click, and then invoke copy (Figure 6).

Figure 6: Copying Series Across EViews Pagefiles



Step 3: Make "Quarterly" the active pagefile, right click on the workfile page, and invoke "Paste Special…" (Figure 7)

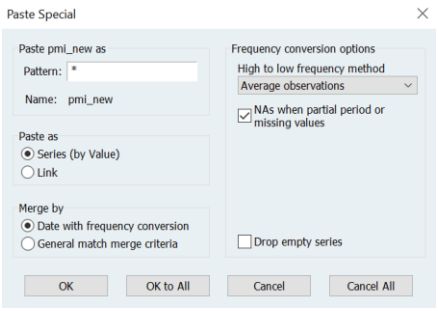
Figure 7: Moving to Quarterly Pagefile and invoking "Paste Special…"



Step 4: Make **pmi_new** the target variable and select "Average observations" for the "High to low frequency method" (Figure 8). EViews will average the monthly values associated with each quarter to

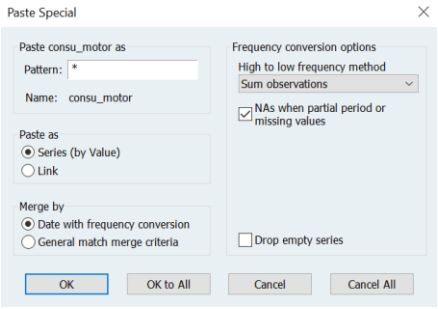
derive the quarterly series. Notice that if any month has a missing value, the associated quarter will also have a missing value ("NAs when partial period or missing values").

Figure 8: Specifying the Conversion Method to a Lower Frequency



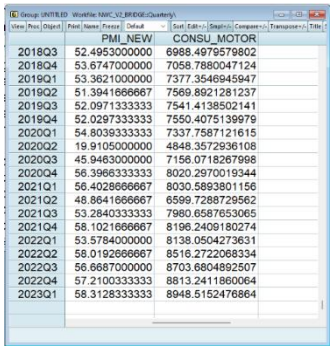
The final step before estimation/forecasting is to repeat Steps 1-4 for the remaining monthly indicators (i.e., **consu_motor**) used in the baseline regression. See the recommended settings in Figure 9.

Figure 9: Specifying the Conversion Method to a Lower Frequency: consu_motor



The result will be two monthly series, **pmi_new** and **consu_motor**, saved in the quarterly pagefile. Both series will have a usable (i.e., non-missing) observation for 2023Q1 that was derived from three non-missing monthly observations (Figure 10).

Figure 10: Quarterly Values of pmi_new and consu_motor



	PMI_NEW	CONSU_MOTOR
2018Q3	52.4953000000	6988.4979579802
2018Q4	53.6747000000	7058.7880047124
2019Q1	53.3621000000	7377.3546945947
2019Q2	51.3941666667	7569.8921281237
2019Q3	52.0971333333	7541.4138502141
2019Q4	52.0297333333	7550.4075139979
2020Q1	54.8039333333	7337.7587121615
2020Q2	19.9105000000	4848.3572936108
2020Q3	45.9463000000	7156.0718267998
2020Q4	56.3966333333	8020.2970019344
2021Q1	56.4028666667	8030.5893801156
2021Q2	48.8641666667	6599.7288729562
2021Q3	53.2840333333	7980.6587653065
2021Q4	58.1021666667	8196.2409180274
2022Q1	53.5784000000	8138.0504273631
2022Q2	58.0192666667	8516.2722068334
2022Q3	56.6687000000	8703.6804892507
2022Q4	57.2100333333	8813.2411860064
2023Q1	58.3128333333	8948.5152476864

Since we want to include a variable to proxy for the impact of Global Financial Crisis, India Demonetization, and COVID-19, we create the CRISIS dummy variable using the following commands under "Command Box":

```
smpl @all
genr CRISIS = 0

smpl 2009Q1 2009Q1
CRISIS = 1

smpl 2016Q4 2016Q4
CRISIS = 1

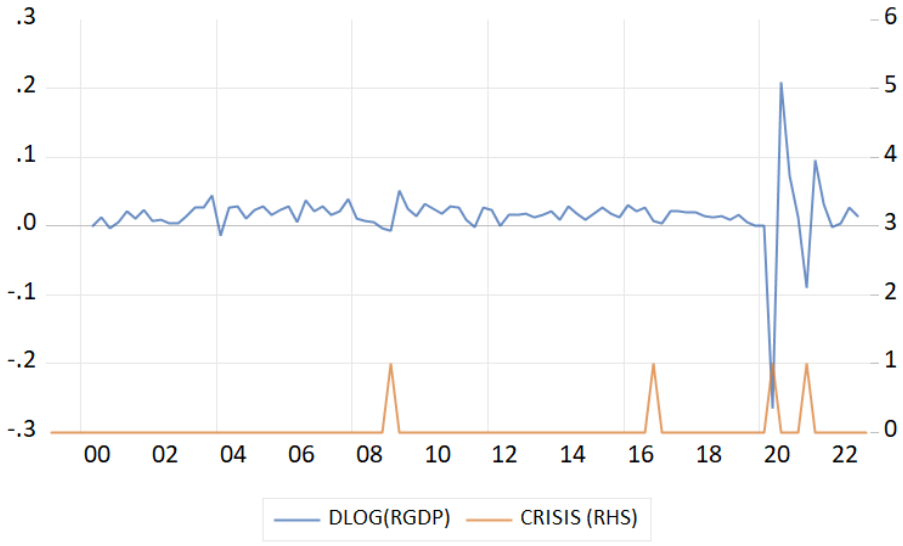
smpl 2020Q2 2020Q2
CRISIS = 1

smpl 2021Q2 2021Q2
CRISIS = 1

smpl @all
```

This dummy variable flags the quarters when the first and second COVID-19 waves might have affected Q-on-Q real GDP growth (i.e., the target variable).

Figure 11: Visualize Q-o-Q growth rate of Real GDP to determine the CRISIS dummy.



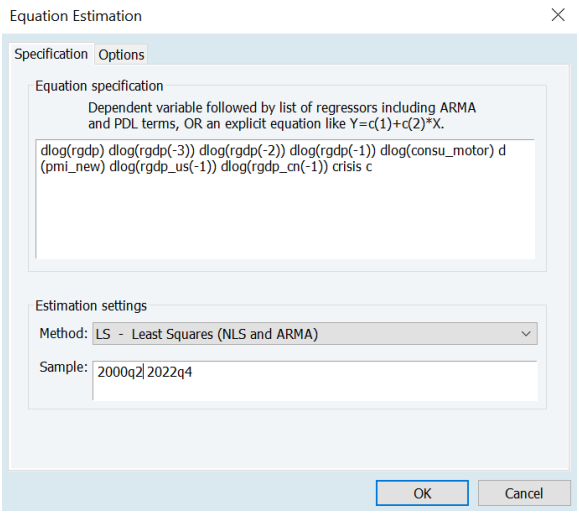
The first step in obtaining a forecast of India's real GDP for **2023Q1** is to estimate the base model using the available data to **2022Q4**.[10]

This is carried out using EViews' ordinary least squares routine, which can be accessed by typing "LS" in the command area. This results in the least squares dialog box appearing on the screen (Figure 12). The dialog box needs to be populated with the specification of the base forecasting regression. Ordinary Least

[10] For simplicity, we ignore the impact of lagged variables on the length of the forecast horizon.

Squares (OLS) estimates of its parameters are obtained by clicking "OK" (Figure 12). Then, save the equation by clicking on "Name" => "**eq_f_base**"

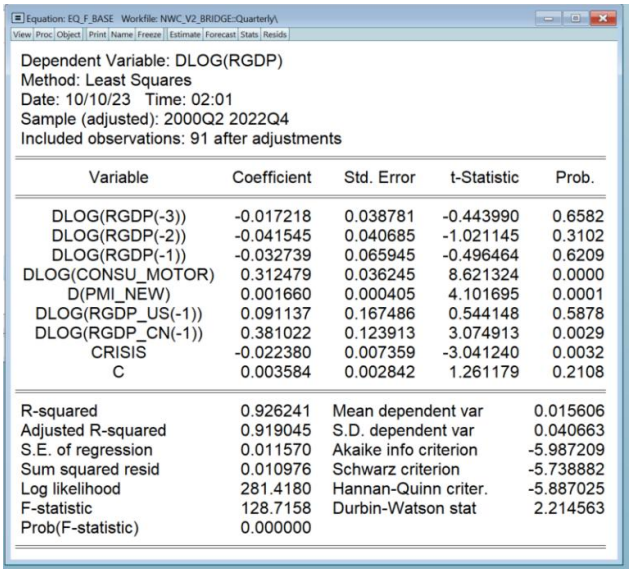
Figure 12: OLS Estimation of the Base Forecasting Model for India



DLOG(RGDP) DLOG(RGDP(-3)) DLOG(RGDP(-2)) DLOG(RGDP(-1)) DLOG(CONSU_MOTOR) D(PMI_NEW) DLOG(RGDP_US(-1)) DLOG(RGDP_CN(-1)) CRISIS C

2000Q2 2022Q4

Figure 13: OLS Results for the Base Forecasting Model for India[11]



Equation: EQ_F_BASE Workfile: NWC_V2_BRIDGE::Quarterly\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: DLOG(RGDP)
Method: Least Squares
Date: 10/10/23 Time: 02:01
Sample (adjusted): 2000Q2 2022Q4
Included observations: 91 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DLOG(RGDP(-3))	-0.017218	0.038781	-0.443990	0.6582
DLOG(RGDP(-2))	-0.041545	0.040685	-1.021145	0.3102
DLOG(RGDP(-1))	-0.032739	0.065945	-0.496464	0.6209
DLOG(CONSU_MOTOR)	0.312479	0.036245	8.621324	0.0000
D(PMI_NEW)	0.001660	0.000405	4.101695	0.0001
DLOG(RGDP_US(-1))	0.091137	0.167486	0.544148	0.5878
DLOG(RGDP_CN(-1))	0.381022	0.123913	3.074913	0.0029
CRISIS	-0.022380	0.007359	-3.041240	0.0032
C	0.003584	0.002842	1.261179	0.2108

R-squared	0.926241	Mean dependent var	0.015606
Adjusted R-squared	0.919045	S.D. dependent var	0.040663
S.E. of regression	0.011570	Akaike info criterion	-5.987209
Sum squared resid	0.010976	Schwarz criterion	-5.738882
Log likelihood	281.4180	Hannan-Quinn criter.	-5.887025
F-statistic	128.7158	Durbin-Watson stat	2.214563
Prob(F-statistic)	0.000000		

Some important remarks about selecting the baseline model:

1. As mentioned in Lecture 1, the baseline nowcasting regression is a "reduced form" model rather than a structural equation. It follows that we cannot use the baseline model to assess the impact of policy changes on the economy.

[11] You may test the serial correlation of the equation by clicking "View" => "Residual Diagnostic" => "Serial Correlation LM test" at 6 lags

2. All the RHS regressors included in the baseline model should meet the following conditions:

    a) Stationarity (i.e., they should not possess a unit root).
    b) No seasonality (i.e., add appropriate dummies to the baseline regression to correct for non-seasonality or seasonally adjust the series using the X-13 procedure before running the regression).
    c) The signs of regressors' estimated coefficients should be consistent with economic priors.

3. The objective of the nowcasting is to minimize RMSE. Thus, it is okay to keep insignificant regressors in the baseline model if they decrease the RMSE and have the correct sign (see condition 2).

Now click on the **Forecast** tab to generate a dynamic forecast for real GDP for **2023Q1** (please ensure that you complete the dialog boxes as indicated below in Figure 14 before clicking on the OK button).

The forecast will be saved to **rgdp_f_2023Q1**, and its standard errors will be saved to **rgdp_f_2023Q1_se**. Click **OK** to generate these series and obtain a graph of actual real GDP against the forecast, together with summary measures of forecasting performance.

Note: for the "Dynamic forecast" option, EViews will use the forecasted values of the lagged dependent variables rather than the actual values. Hence, forecast errors will tend to accumulate the longer the forecast horizon is. It follows that the starting date of the "Forecast" sample is critical for the forecast evaluation result. Since we only want to find the back-test result for 2023Q1, we may only include just one period in the forecast sample (as below):

Figure 14: Forecasting 2023Q1 Real GDP

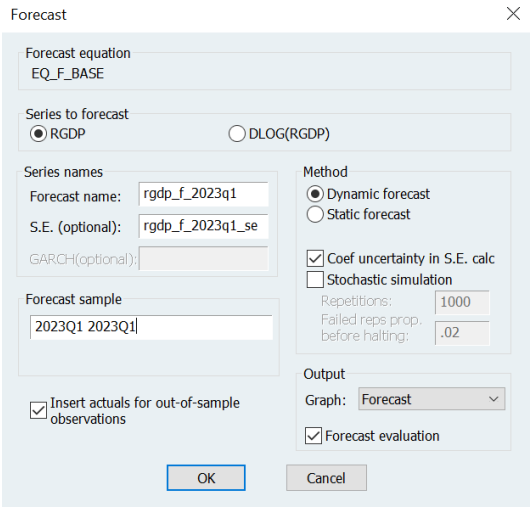
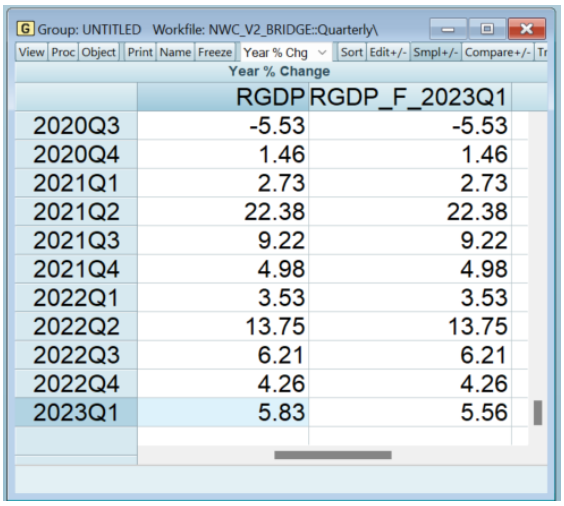


Figure 15: Forecast for Real GDP, India, 2023Q1 (YoY Growth Rate)


Group: UNTITLED Workfile: NWC_V2_BRIDGE::Quarterly\

View | Proc | Object | Print | Name | Freeze | Year % Chg | Sort | Edit+/- | Smpl+/- | Compare+/- | Tr

Year % Change

	RGDP	RGDP_F_2023Q1
2020Q3	-5.53	-5.53
2020Q4	1.46	1.46
2021Q1	2.73	2.73
2021Q2	22.38	22.38
2021Q3	9.22	9.22
2021Q4	4.98	4.98
2022Q1	3.53	3.53
2022Q2	13.75	13.75
2022Q3	6.21	6.21
2022Q4	4.26	4.26
2023Q1	5.83	5.56

Figure 15 provides the actual values of real GDP and its forecast for 2023Q1 using the baseline model. With all the available information from high-frequency indicators, we predict the Y-o-Y real GDP growth rate to be 5.56 percent, whereas the actual growth rate is 5.83 percent.

Section III: Bridge Estimation

The next step is to nowcast real GDP to **2023Q2** using the Bridge approach.[12]

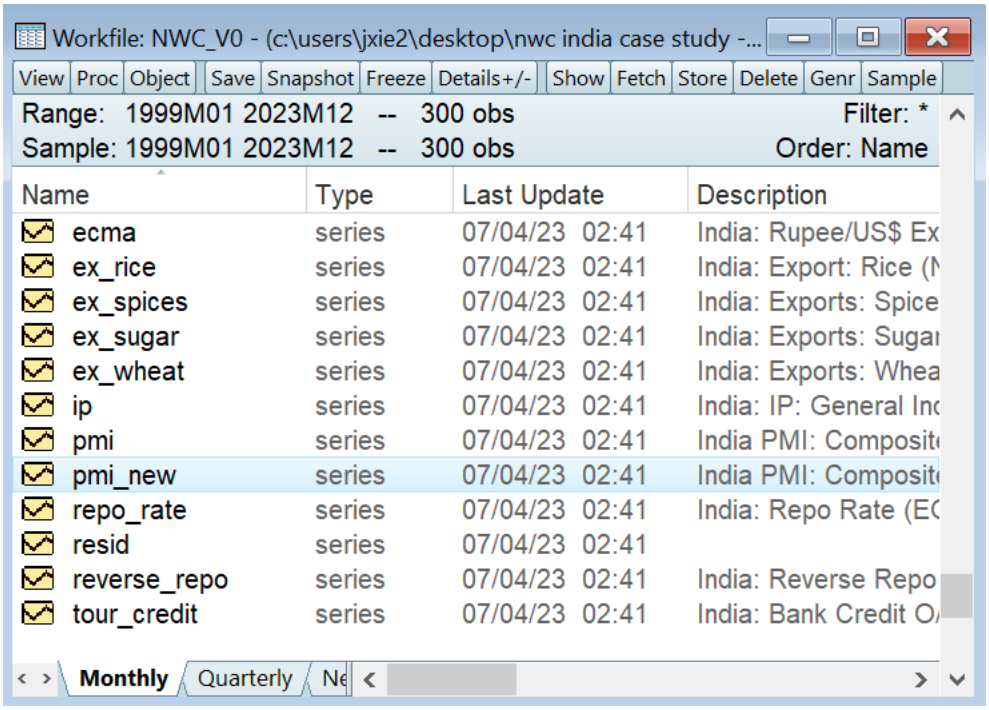
Recall that all monthly indicators are available until 2023M05, and they need to be extended to 2023M06 to nowcast real GDP to **2023Q2**. Both quarterly indicators are available until 2023Q1. Since we are using 1 lag for each quarterly series, no extension is needed for these indicators as they already have valid data for 2023Q2.

Extending the **two independent variables** to have valid data for the target frequency is the essence of the Bridge approach to nowcasting. Simply put, we need forecasts for **pmi_new** and **consu_motor** for 2023M06. Any forecasting method can be used, including automatic ARIMA procedures, which EViews supports. We will demonstrate the use of the automatic ARIMA approach using **pmi_new**.

Step 1: Make the pagefile "Monthly" the active pagefile and select "**pmi_new**" (Figure 16).

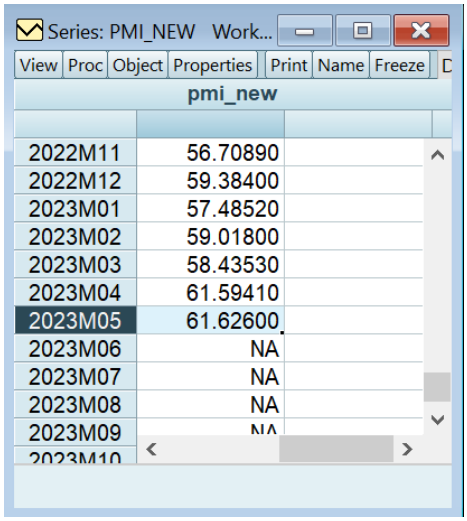
[12] See **2_bridge.prg** for the command line equivalents of the estimation/forecasting procedures used in this section.

Figure 16: Monthly Page File, India Database



Double-click **pmi_new** to see its monthly values. Notice that the series ends in 2023M05 (Figure 17).

Figure 17: Actual Values of pmi_new



The next step is to invoke "**Automatic ARIMA Forecasting**" from the "Proc" menu tab. Doing so yields the dialog box shown in Figure 18.

Figure 18: Invoking Automatic Box-Jenkin's Forecasting in EViews

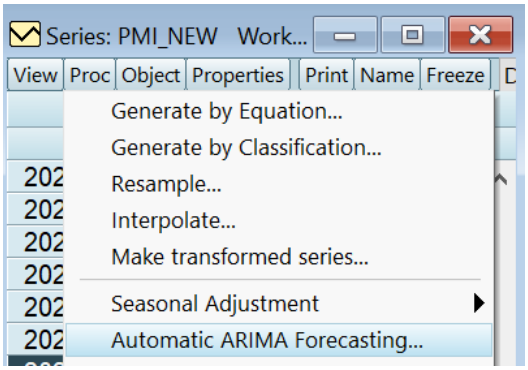
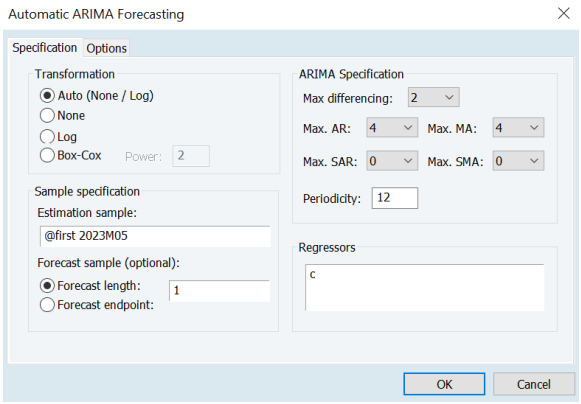


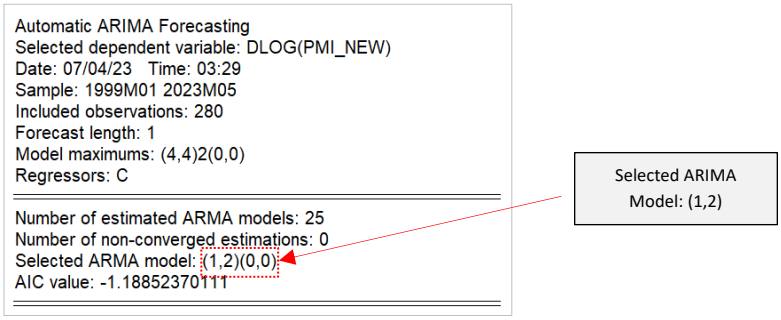
Figure 19: Invoking Automatic Box-Jenkin's Forecasting in EViews (pmi_new)



Using the AIC criterion, EViews will find the "best" Box-Jenkins ARIMA model for the period 2000M01 ~ 2023M05 across the 25 models estimated (as implied by the settings for maximum AR and maximum MA) and then use the preferred model to forecast **pmi_new** for one month (i.e., the forecast length) out to **2023M06**.

The default selection criterion is the Akaike Information Criterion (click the Options tab to see this), which is appropriate for selecting an out-of-sample forecasting model. The result is shown in Figure 20:

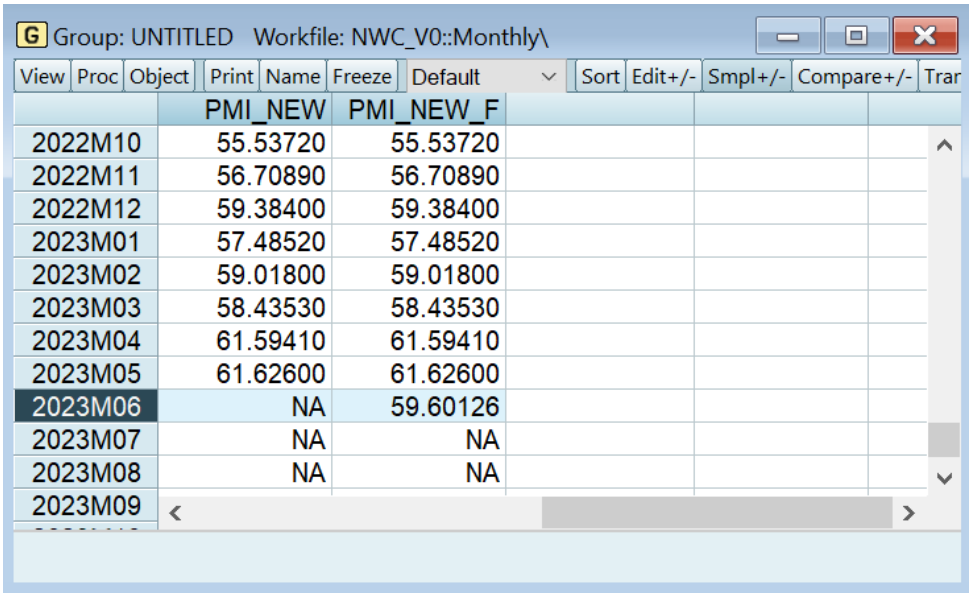
Figure 20: Automatic Box-Jenkins Forecasts for pmi_new



According to the Akaike Information Criterion (AIC), the preferred forecasting model for **pmi_new** is an ARMA (1,2) model (i.e., 1 auto-regressive lag, and 2 moving average terms).

Once the preferred model is identified, we use it to extend the series out to 2023M06. The actual of **pmi_new** and the forecast values of **pmi_new_f** are shown in Figure 21.

Figure 21: Automatic Box-Jenkins Forecasts for pmi_new: result



	PMI_NEW	PMI_NEW_F
2022M10	55.53720	55.53720
2022M11	56.70890	56.70890
2022M12	59.38400	59.38400
2023M01	57.48520	57.48520
2023M02	59.01800	59.01800
2023M03	58.43530	58.43530
2023M04	61.59410	61.59410
2023M05	61.62600	61.62600
2023M06	NA	59.60126
2023M07	NA	NA
2023M08	NA	NA
2023M09		

The next step is to extend the monthly version of **consu_motor** to 2023M06 using the ARIMA forecasting function (invoked while in the "**Monthly**" pagefile).

Figure 22: Invoking Automatic Box-Jenkin's Forecasting in EViews (consu_motor)

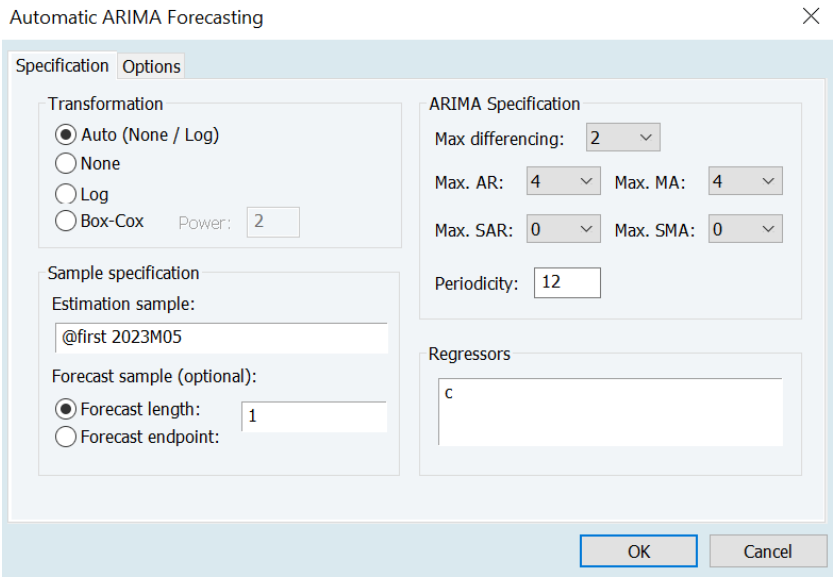
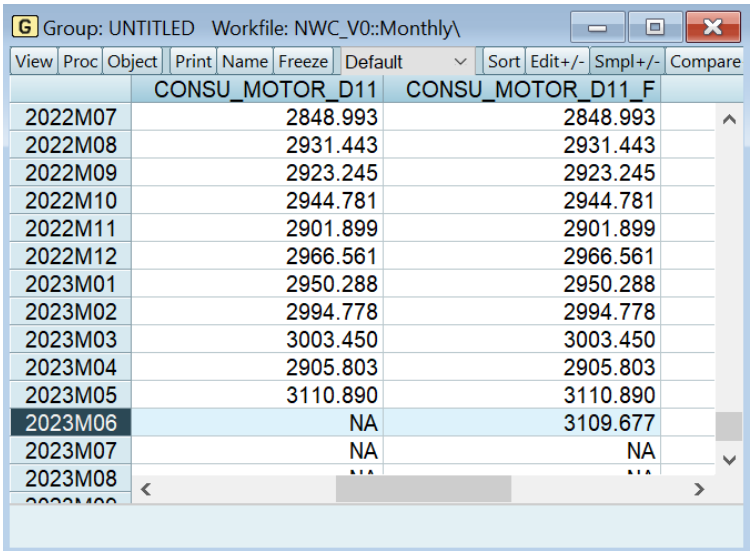


Figure 23: Automatic Box-Jenkin's Forecasting in EViews (Result: consu_motor_d11)

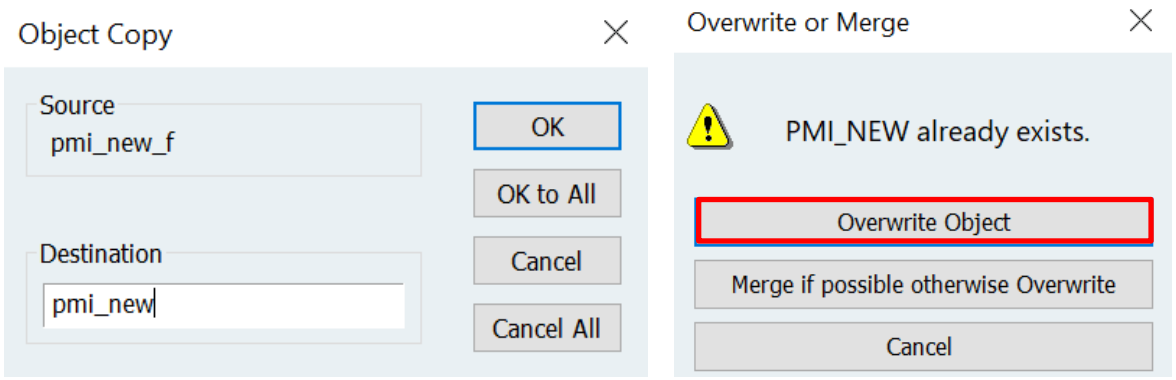


	CONSU_MOTOR_D11	CONSU_MOTOR_D11_F
2022M07	2848.993	2848.993
2022M08	2931.443	2931.443
2022M09	2923.245	2923.245
2022M10	2944.781	2944.781
2022M11	2901.899	2901.899
2022M12	2966.561	2966.561
2023M01	2950.288	2950.288
2023M02	2994.778	2994.778
2023M03	3003.450	3003.450
2023M04	2905.803	2905.803
2023M05	3110.890	3110.890
2023M06	NA	3109.677
2023M07	NA	NA
2023M08		

Now that we have extended all monthly indicators to 2023M06, we need to merge the forecasted series with the original series. This step can be completed by copying the forecasted series (all of which end with "_f") and paste it with "_f" removed from the name.

The procedure is demonstrated for **pmi_new_f** in Figure 24. Repeat the same procedure for **consu_motor_d11_f**.

Figure 24: Merge Forecasted Series with Original Series



We now need to convert those updated monthly series to the quarterly frequency. To keep output well organized, we will create a quarterly "Bridge" pagefile to file these series. Issue the following EViews commands create the quarterly Bridge pagefile:

```
pageselect quarterly
pagecopy(page=Bridge)
```

Press "OK" if a dialog box pops up when you issue these commands.

Simply repeat the "copy and paste special" activity to merge the forecasted monthly frequency (i.e., **pmi_new and consu_motor**) to the quarterly Bridge pagefile.

Figure 25: Quarterly Forecast for pmi_new and consu_motor

Group: UNTITLED Workfile: NWC_V2_BRIDGE::Quarterly\

View | Proc | Object | Print | Name | Freeze | Default | Sort | Edit+/- | Smpl+/- | Compare+/- | Transpose+/-

	PMI_NEW	CONSU_M...
2021Q4	58.10217	8196.241
2022Q1	53.57840	8138.050
2022Q2	58.01927	8516.272
2022Q3	56.66870	8703.680
2022Q4	57.21003	8813.241
2023Q1	58.31283	8948.515
2023Q2	60.94046	9126.371
2023Q3	NA	NA
2023Q4	NA	NA

Starting from the estimates of the base equation (i.e., **eq_f_base**) for 2000Q2-2022Q4, we will re-estimate the equation until **2023Q1** (naming it **eq01_f_bridge**).

Figure 26: Bridge Equation

Equation: EQ01_F_BRIDGE Workfile: NWC_V2_BRIDGE::Quarterly\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: DLOG(RGDP)
Method: Least Squares
Date: 10/10/23 Time: 02:01
Sample (adjusted): 2000Q2 2023Q1
Included observations: 92 after adjustments

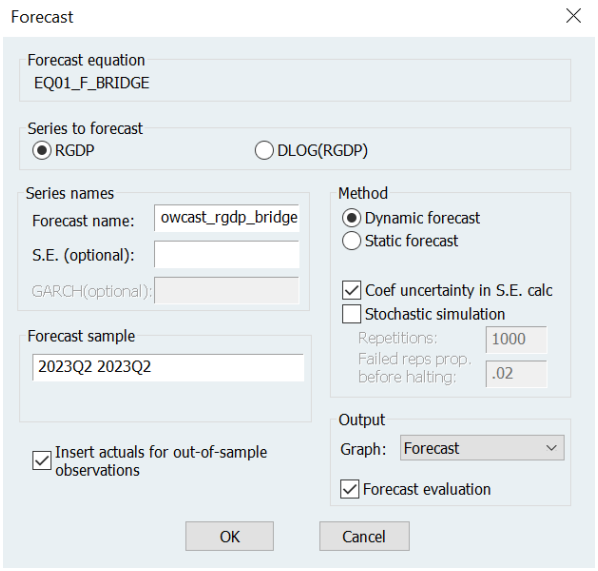
Variable	Coefficient	Std. Error	t-Statistic	Prob.
DLOG(RGDP(-3))	-0.017240	0.038557	-0.447114	0.6560
DLOG(RGDP(-2))	-0.040927	0.040348	-1.014354	0.3134
DLOG(RGDP(-1))	-0.032954	0.065557	-0.502680	0.6165
DLOG(CONSU_MOTOR)	0.312822	0.036000	8.689424	0.0000
D(PMI_NEW)	0.001671	0.000400	4.180365	0.0001
DLOG(RGDP_US(-1))	0.091918	0.166480	0.552128	0.5823
DLOG(RGDP_CN(-1))	0.376061	0.121011	3.107649	0.0026
CRISIS	-0.022421	0.007314	-3.065602	0.0029
C	0.003694	0.002779	1.329025	0.1875

R-squared	0.926203	Mean dependent var	0.015577
Adjusted R-squared	0.919090	S.D. dependent var	0.040440
S.E. of regression	0.011503	Akaike info criterion	-5.999733
Sum squared resid	0.010982	Schwarz criterion	-5.753036
Log likelihood	284.9877	Hannan-Quinn criter.	-5.900164
F-statistic	130.2139	Durbin-Watson stat	2.221487
Prob(F-statistic)	0.000000		

Click on the **Forecast** tab and complete the required options as shown in Figure 27. Click the **OK** button to obtain the results, and the forecast for quarterly GDP shown in Figure 28.

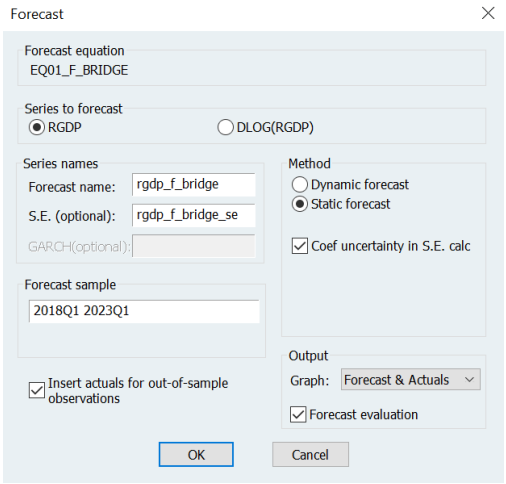
Note that the quarterly series for real GDP has been extended to **2023Q2**, as required. The nowcast result can be found in the variable called as "**nowcast_rgdp_bridge**".

Figure 27: Bridge forecast for Real GDP of India



We can also conduct an in-sample evaluation to assess the forecasting performance of the baseline model by repeating the same proceudre with the forecast sample set to "**2018Q1 ~ 2023Q2**". In this way, we can compare the bridge forecasted series with the actual series. We will name this series "**rgdp_f_bridge**".

Figure 28: In-Sample Evaluation: Bridge, 2018Q1~2023Q2



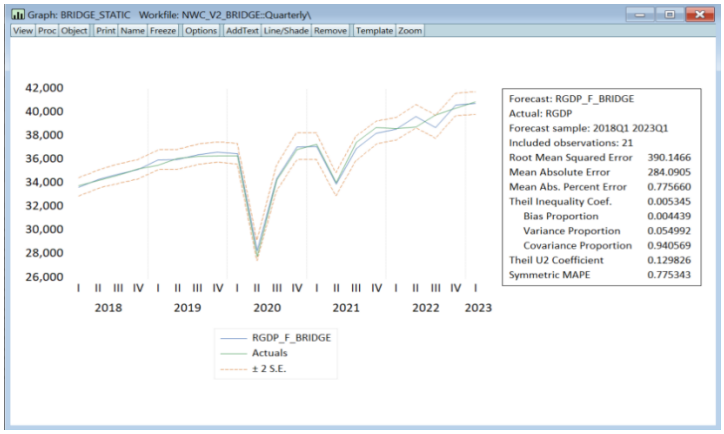
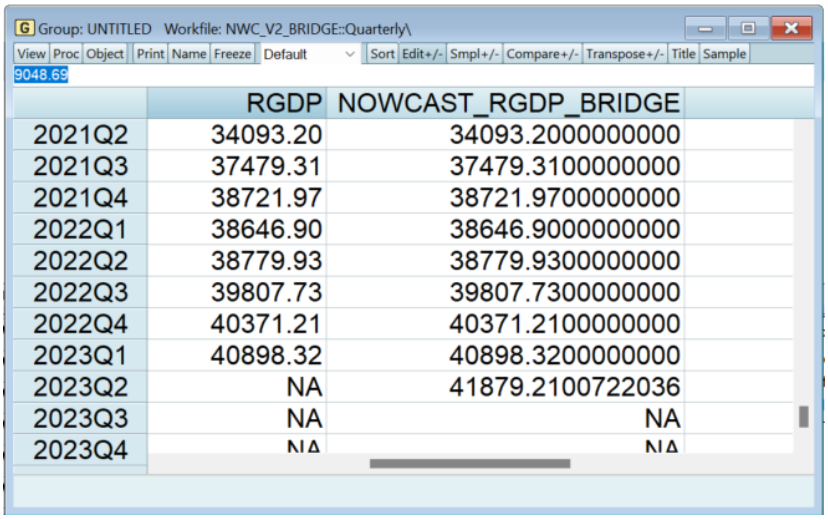


Figure 29: Actual and Nowcasted Values of Real GDPR



	RGDP	NOWCAST_RGDP_BRIDGE
2021Q2	34093.20	34093.2000000000
2021Q3	37479.31	37479.3100000000
2021Q4	38721.97	38721.9700000000
2022Q1	38646.90	38646.9000000000
2022Q2	38779.93	38779.9300000000
2022Q3	39807.73	39807.7300000000
2022Q4	40371.21	40371.2100000000
2023Q1	40898.32	40898.3200000000
2023Q2	NA	41879.2100722036
2023Q3	NA	NA
2023Q4	NA	NA

Section IV: Nowcasting using the MIDAS Approach

Similar to Bridge, we will first create a quarterly "Midas" pagefile to store the results from the Midas/U-MIDAS estimation.

```
pageselect Bridge
pagecopy(page=Midas)
```

The base model remains the same as in bridge estimation. MIDAS differs from the Bridge estimator only in the manner in which it incorporates the high frequency variables into the base regression.[13] The fundamental aim of MIDAS estimation is to reduce the number of parameters to estimate in the base frequency regression, which can be large owing to the use of split sampling (see below).

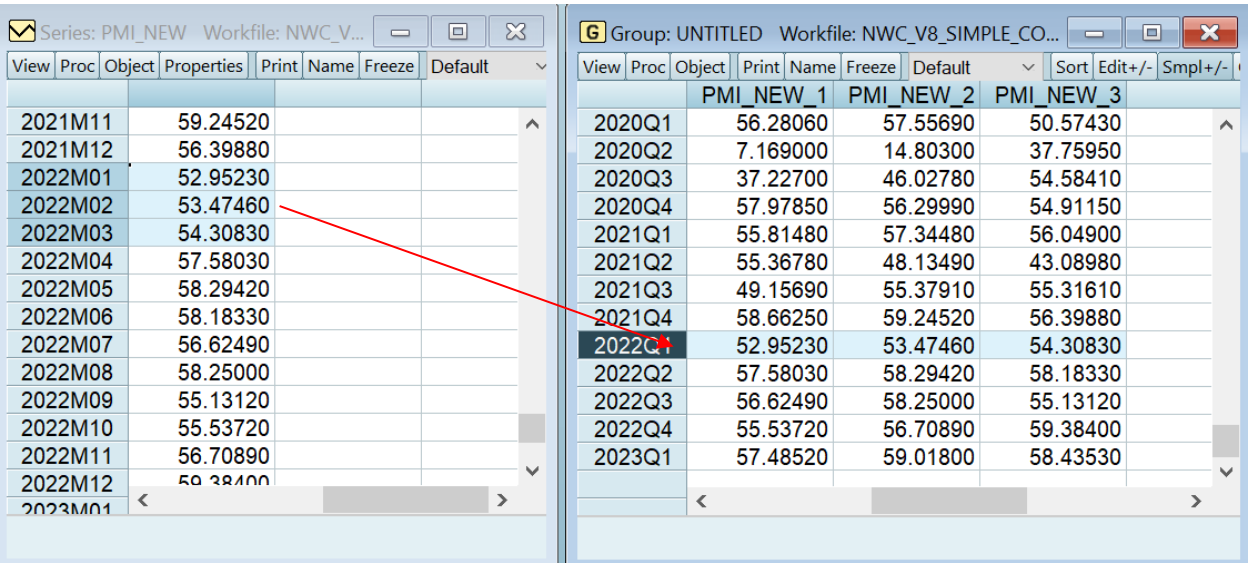
To understand the MIDAS approach intuitively, consider the monthly series **pmi_new**, which spans 2000M01~2023M06 for India (including forecasted values for 2023M06). For a given quarter, we can choose any one of the (three) monthly values associated with that quarter to represent the quarterly value of **pmi_new**. For 2022, for example, we could create a quarterly series from the monthly series by selecting the January, April, July, and October values of the monthly observations (and so on for each subsequent year). Or we could create a separate quarterly series by starting with the February monthly value, followed by the May, August, and November values. We could also define a third quarterly series by selecting the March, June, September, and December values (and so on for each subsequent year). Thus, a monthly series will yield three quarterly variables to add to the base forecast regression.

To make this procedure more concrete, Figure 30 shows the three quarterly series that are associated with the monthly **pmi_new** variable. Notice that the value for 2022Q1 can be 52.9, 53.5, and 54.3, depending

[13] See midas.prg (first half) for the command line equivalents of the estimation/forecasting procedures used in this section.

on which monthly observation we use for the first quarter (the actual monthly values of **pmi_new** are shown in the first screenshot).

Figure 30: Quarterly Series that can be Produced from the Monthly pmi_new



This approach of generating low-frequency variables from higher frequency variables is called "skip" or "split" sampling.[14]

Depending on the number of high-frequency values in the target (lower) frequency, split sampling can produce a significant number of low-frequency variables to add to the base regression, resulting in a large number of parameters to estimate before the forecasts can be produced. For example, given that there are 90-92 days in a given quarter, split-sampling a daily indicator would yield 90-92 variables to add to the base forecasting, substantially increasing the number of parameters in the baseline (quarterly) model to estimate.

MIDAS tackles this problem by assuming that the unknown parameters on the split-sampled series follow a specific functional form that depends on far fewer parameters than the number of high-frequency variables in the regression, e.g., a polynomial distributed lag function (see the lecture notes).

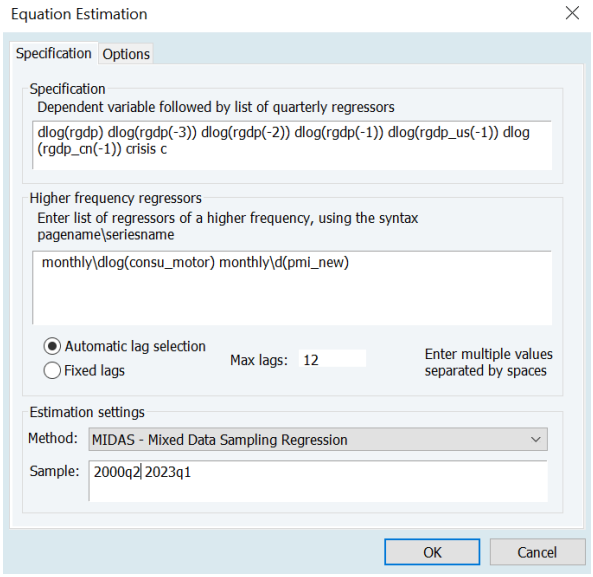
Implementing MIDAS is straight-forward using EViews. The baseline regression with the high-frequency indicators **pmi_new** and **consu_motor** is estimated using the LS command and its MIDAS option (Figure 31).[15] The base model allows for up to 12 lags (including the 0 lag) of each split-sampled high-frequency indicator, with the optimal number chosen using the Akaike Information Criterion. The parameter estimates on the 12 lags are assumed to follow an Almon PDL function of order 3, which depends only on 3 unknown parameters (One can specify the order of the Almon PDL function using the Options tab).

[14] You may create the "split" variable by click "copy" => "paste special" => "Split Observation" => "Split Index (1~3 for monthly to quarterly conversion)".

[15] The dialog box shown in Figure 31, without any pre-filled specification or high frequency regressors, can be obtained by typing "MIDAS" in the command area. Alternatively, you can type "LS" and select MIDAS for the estimation method.

*Save the equation by clicking on "Name" => "**eq02_f_midas**".*

Figure 31: Estimating a MIDAS Model Using EViews



dlog(rgdp) dlog(rgdp(-3)) dlog(rgdp(-2)) dlog(rgdp(-1)) dlog(rgdp_us(-1)) dlog(rgdp_cn(-1)) crisis c

monthly\dlog(consu_motor) monthly\d(pmi_new)

2000q2 2023q1

Figure 32: MIDAS Results for Base Regression, India

Dependent Variable: DLOG(RGDP)
Method: MIDAS
Date: 07/06/23 Time: 21:56
Sample (adjusted): 2001Q1 2023Q1
Included observations: 89 after adjustments
Method: PDL/Almon (polynomial degree: 3)
Automatic lag selection, max lags: 12
Chosen selection: 7 4

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DLOG(RGDP(-3))	-0.020142	0.043859	-0.459241	0.6474
DLOG(RGDP(-2))	-0.066520	0.048344	-1.375972	0.1729
DLOG(RGDP(-1))	-0.298598	0.078225	-3.817192	0.0003
DLOG(RGDP_US(-1))	0.322217	0.173683	1.855200	0.0674
DLOG(RGDP_CN(-1))	0.475069	0.120698	3.936023	0.0002
CRISIS	-0.018463	0.007657	-2.411244	0.0183
C	0.003064	0.003237	0.946786	0.3467
Page: MONTHLY Series: DLOG(CONSU_MOTOR) Lags: 7				
PDL01	-0.052932	0.035219	-1.502926	0.1370
PDL02	0.165082	0.026152	6.312512	0.0000
PDL03	-0.021069	0.003347	-6.294646	0.0000
Page: MONTHLY Series: D(PMI_NEW) Lags: 4				
PDL01	-0.003007	0.001114	-2.700347	0.0085
PDL02	0.003891	0.001081	3.599417	0.0006
PDL03	-0.000753	0.000220	-3.421110	0.0010

R-squared	0.936516	Mean dependent var	0.015984
Adjusted R-squared	0.926493	S.D. dependent var	0.041041
S.E. of regression	0.011127	Akaike info criterion	-6.024622
Sum squared resid	0.009410	Schwarz criterion	-5.661113
Log likelihood	281.0957	Hannan-Quinn criter.	-5.878102
Durbin-Watson stat	2.085819		

MONTHLY\DLOG(C...	Lag	Coefficient	Distribution
	0	0.091081	
	1	0.192957	
	2	0.252694	
	3	0.270295	
	4	0.245757	
	5	0.179082	
	6	0.070270	

MONTHLY\D(PMI_...	Lag	Coefficient	Distribution
	0	0.000130	
	1	0.001760	
	2	0.001884	
	3	0.000501	

Irrespective of the number of lags chosen or the number of split sampled variables for each high-frequency variable, note that only 3 parameters need to be estimated for each high-frequency indicator. EViews calculates the implied values of these parameters in the output table (Figure 32). In the above example, 11 parameters (4 for **pmi_new**, 7 for **consu_motor**) have been estimated using 6 Almon PDL parameters.

Forecasting using a MIDAS regression proceeds the same as the Bridge estimator. Click on the "**Forecast**" button to generate a dynamic nowcast for 2023Q2 and a static forecast for 2018Q1-2023Q1 (Figures 33-35). The nowcast series will be saved as "**nowcast_rgdp_midas**" and the forecast will be saved in a series called "**rgdp_f_midas"**.

Figure 33: Nowcast & Forecast Dialog Box for a MIDAS Regression

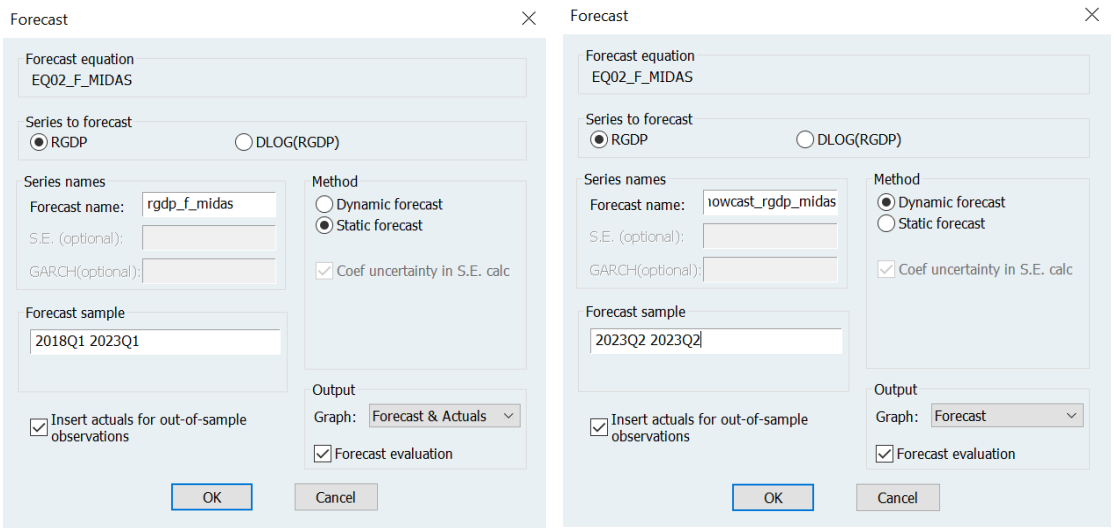


Figure 34: Forecast Graph and Forecast Evaluation Table, 2018Q1~2023Q1

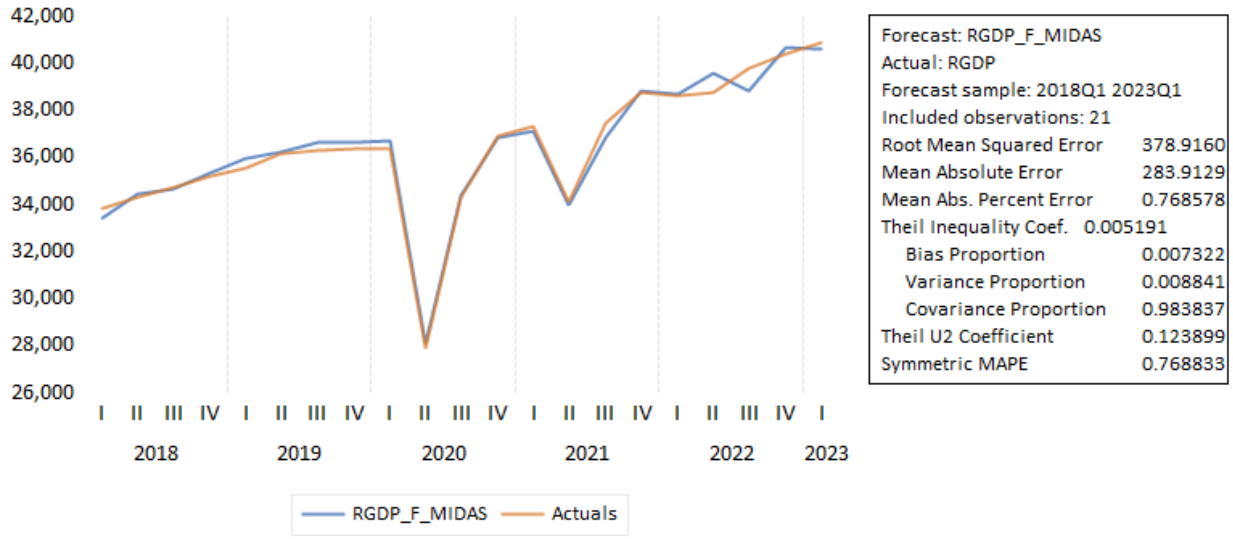


Figure 35: Actual and Nowcasted Values for Real GDP Using MIDAS, India

Group: UNTITLED Workfile: NWC_V3_MIDAS::Quarterly\

View | Proc | Object | Print | Name | Freeze | Default | Sort | Edit+/- | Smpl+/- | Compare+/- | Transpose+

	RGDP	NOWCAST_RGDP_MIDAS
2021Q3	37479.31	37479.31
2021Q4	38721.97	38721.97
2022Q1	38646.90	38646.90
2022Q2	38779.93	38779.93
2022Q3	39807.73	39807.73
2022Q4	40371.21	40371.21
2023Q1	40898.32	40898.32
2023Q2	NA	42035.13
2023Q3	NA	NA
2023Q4	NA	NA

Section V: Nowcasting using the U-MIDAS Approach

U-MIDAS is simply unconstrained MIDAS.[16] That is, we add the split-sampled variables to the base regression and estimate it "as-is" without making any assumptions about the shape of the unknown parameters in the baseline regression. We do not estimate using PDL/Almon lags or any other method designed to reduce the number of parameters to estimate.

[16] See midas.prg for the command line equivalents of the estimation/forecasting procedures used in this section.

U-MIDAS is especially useful when the frequency mismatch between the target (low) frequency and the high-frequency indicators is small. This is the case for India, where the target frequency is quarterly, and all the high-frequency indicator variables have monthly frequencies.

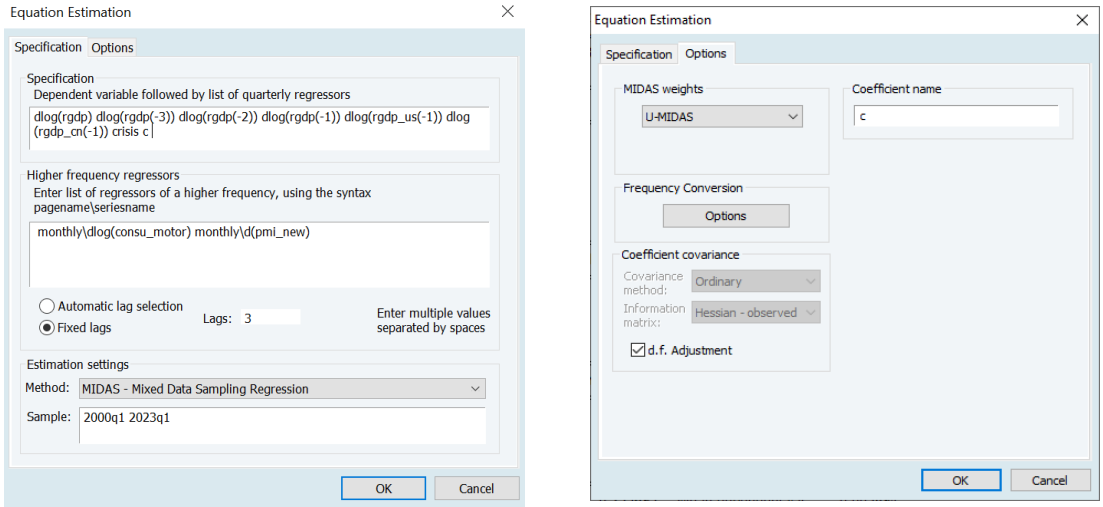
Estimation of a U-MIDAS model in EViews proceeds as shown in Figure 36, and U-MIDAS is selected as the method for including the high-frequency indicators. Then save the equation by clicking on "Name" => "**eq03_f_umidas**".

```
dlog(rgdp) dlog(rgdp(-3)) dlog(rgdp(-2))
dlog(rgdp(-1)) dlog(rgdp_us(-1))
dlog(rgdp_cn(-1)) crisis c

monthly\dlog(pmi_new)
monthly\dlog(consu_motor)

2000q2 2023q1
```

Figure 36: U-MIDAS Estimation in EViews



Press **OK** to obtain the estimated parameters of the base forecasting model (Figure 37):

Figure 37: U-MIDAS estimation with EViews

Dependent Variable: DLOG(RGDP)
Method: MIDAS
Date: 07/06/23 Time: 21:56
Sample (adjusted): 2000Q2 2023Q1
Included observations: 92 after adjustments
Method: U-MIDAS

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DLOG(RGDP(-3))	-0.230577	0.047897	-4.814000	0.0000
DLOG(RGDP(-2))	-0.329657	0.045581	-7.232417	0.0000
DLOG(RGDP(-1))	-0.353285	0.078012	-4.528589	0.0000
DLOG(RGDP_US(-1))	-0.019362	0.195930	-0.098822	0.9215
DLOG(RGDP_CN(-1))	0.659902	0.131217	5.029081	0.0000
CRISIS	-0.024686	0.008879	-2.780212	0.0068
C	0.017987	0.004340	4.144353	0.0001
Page: MONTHLY Series: DLOG(CONSU_MOTOR) Lags: 3				
LAG1	0.032945	0.029170	1.129415	0.2621
LAG2	0.034317	0.040560	0.846094	0.4001
LAG3	0.103788	0.031891	3.254443	0.0017
Page: MONTHLY Series: D(PMI_NEW) Lags: 3				
LAG1	-8.40E-05	0.000539	-0.155918	0.8765
LAG2	0.000458	0.000732	0.626570	0.5327
LAG3	0.002978	0.000613	4.855932	0.0000

R-squared	0.904865	Mean dependent var	0.015577
Adjusted R-squared	0.890414	S.D. dependent var	0.040440
S.E. of regression	0.013387	Akaike info criterion	-5.658794
Sum squared resid	0.014158	Schwarz criterion	-5.302454
Log likelihood	273.3045	Hannan-Quinn criter.	-5.514972
Durbin-Watson stat	1.611446		

The base setting for the number of lags for the high frequency indicators is 3, and that is exactly the number of parameters that are estimated for each high frequency indicator. Thus, unlike MIDAS, U-MIDAS does not impose any constraints on the shape of the lagged parameter space.

Forecasting using a U-MIDAS regression proceeds as before. Click on the "Forecast" tab to generate dynamic nowcast for 2023Q2 (**nowcast_rgdp_umidas**) and a static forecast for 2018Q1-2023Q1 (**rgdp_f_umidas**). See Figure 38.

Figure 38: Forecast Dialog Box for a U-MIDAS Regression

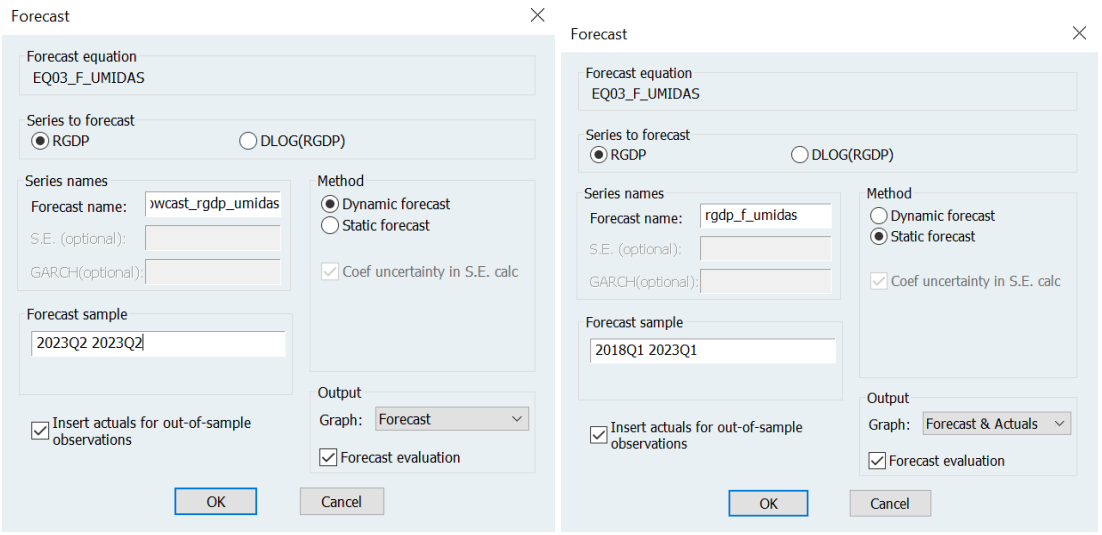


Figure 39: Forecast Graph and Forecast Evaluation Table, U-MIDAS

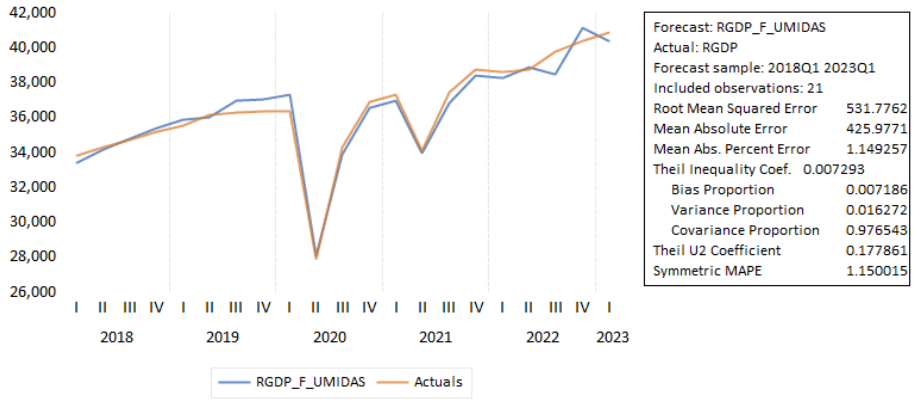
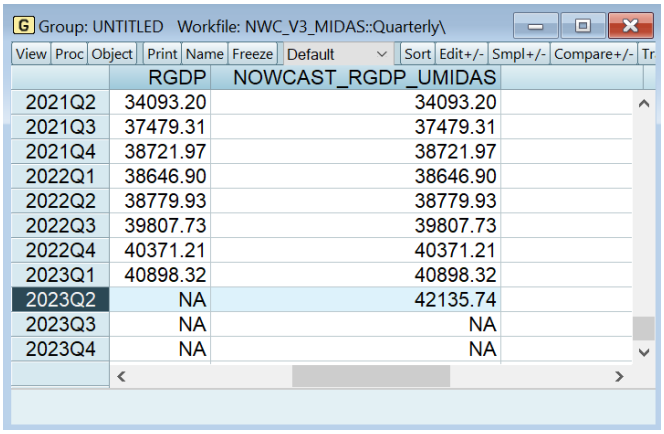


Figure 40: Actual and Nowcasted Values for Real GDP Using U-MIDAS, India



	RGDP	NOWCAST_RGDP_UMIDAS
2021Q2	34093.20	34093.20
2021Q3	37479.31	37479.31
2021Q4	38721.97	38721.97
2022Q1	38646.90	38646.90
2022Q2	38779.93	38779.93
2022Q3	39807.73	39807.73
2022Q4	40371.21	40371.21
2023Q1	40898.32	40898.32
2023Q2	NA	42135.74
2023Q3	NA	NA
2023Q4	NA	NA

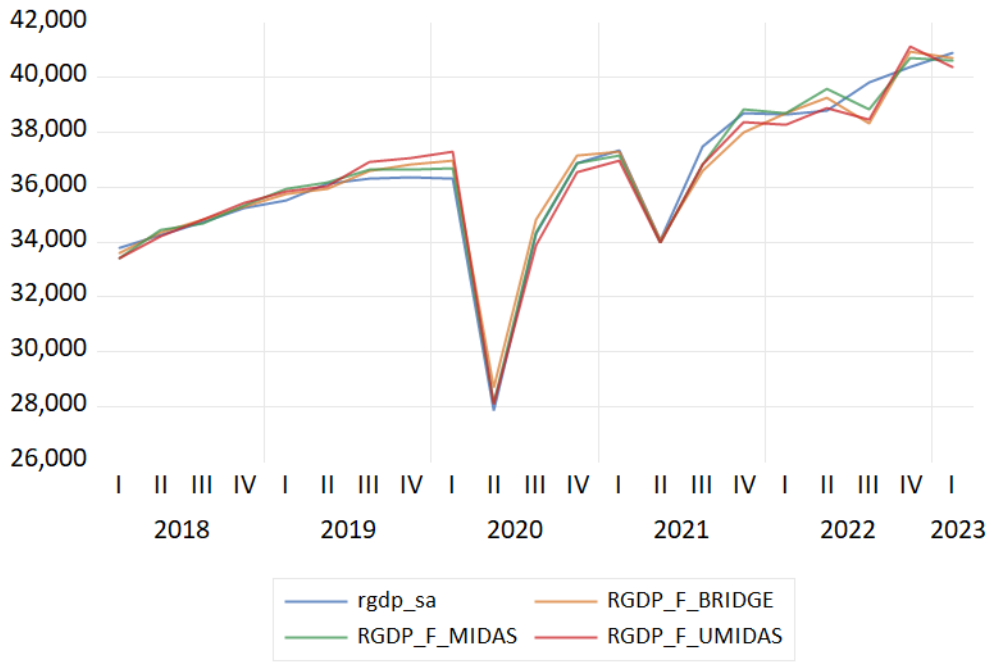
Section VI: Recap: Nowcasting using Bridge, MIDAS, and U-MIDAS

The three methods available for nowcasting are Bridge, MIDAS and U-MIDAS. Irrespective of the method used, one needs to decide what the target frequency will be (e.g., quarterly for the real GDP of India).[17] Bridge produces a forecast by converting all the high-frequency indicators to the target frequency. The high frequency indicators must be extended to encompass the required terminal forecast date.

Under MIDAS and U-MIDAS, the high frequency variables are converted to the target frequency using "skip" or "split" sampling (rather than averaging or summing, for example, all the high frequency values associated with a lower frequency period). These variables are added to the base forecasting model directly under U-MIDAS without any assumptions regarding the shape of their associated parameters. With MIDAS, however, the parameters on the skip-sampled variables are assumed to follow a specific functional form that depends on fewer parameters, thereby reducing the overall number of parameters to estimate.

Figure 41 shows the separate forecasts for **2018Q1-2023Q1** arising from the four forecasts generated above.[18]

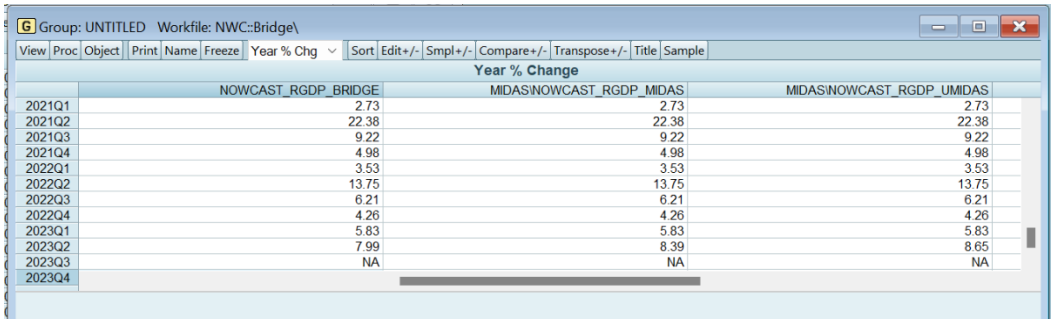
Figure 41: Quarterly Real GDP, and 2018Q1-2023Q1 in-sample evaluation for Bridge, MIDAS and U-MIDAS models



[17] Using the annual frequency would also have been an option for India. The high frequency variables could then be selected from the quarterly and monthly pagefiles.

[18] You may create the graph by first group the forecasted rgdp series and then "View" => "Graph".

Figure 42: Nowcasting Results from Bridge, MIDAS and UMIDAS Models, 2023Q2 (YoY Growth Rate)



Group: UNTITLED Workfile: NWC::Bridge\

View | Proc | Object | Print | Name | Freeze | Year % Chg | Sort | Edit+/- | Smpl+/- | Compare+/- | Transpose+/- | Title | Sample

Year % Change

	NOWCAST_RGDP_BRIDGE	MIDAS\NOWCAST_RGDP_MIDAS	MIDAS\NOWCAST_RGDP_UMIDAS
2021Q1	2.73	2.73	2.73
2021Q2	22.38	22.38	22.38
2021Q3	9.22	9.22	9.22
2021Q4	4.98	4.98	4.98
2022Q1	3.53	3.53	3.53
2022Q2	13.75	13.75	13.75
2022Q3	6.21	6.21	6.21
2022Q4	4.26	4.26	4.26
2023Q1	5.83	5.83	5.83
2023Q2	7.99	8.39	8.65
2023Q3	NA	NA	NA
2023Q4			

You may create the combination forecast for these three nowcasts using weighted or simple average (We will explained forecast combination in detail in Workshop 9).

For example, you may create the simple average pooling forecast for BRIDGE, MIDAS, and U-MIDAS using the following command lines:

```
smpl 2018Q1 2023Q1

rgdp.forceval(mean) bridge\rgdp_f_bridge Midas\rgdp_f_midas Midas\rgdp_f_umidas
```

Figure 43: Pooling Forecast for BRIDGE, MIDAS, and U-MIDAS

Evaluation statistics

Forecast	RMSE	MAE	MAPE	SMAPE	Theil U1	Theil U2
BRIDGE\RGDP_F_BR...	390.1466	284.0905	0.775660	0.775343	0.005345	0.129826
RGDP_F_MIDAS	378.9160	283.9129	0.768578	0.768833	0.005191	0.123899
RGDP_F_UMIDAS	531.7762	425.9771	1.149257	1.150015	0.007293	0.177861
Simple mean	399.3466	306.8167	0.830124	0.830664	0.005473	0.131731

Appendix A: EViews Command Line Approach

As shown above, EViews offers an intuitive, menu-based approach to generating forecasts/nowcasts. The same results can be generated using EViews' programming language and estimation/forecasting objects.

The EViews codes to replicate the workshop results presented above are in the following program files: "*2_bridge.prg*" and "*3_midas.prg*".

